Kilian Shi

Seeking Summer 2027 Internship | Available May 12 – Aug 17
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EDUCATION

Carnegie Mellon University May 2028
Master of Science in Mobile and IoT Engineering

Chongqing University Jul 2026
Bachelor of Engineering in Computer Science and Technology *GPA: 88/100*

PROFESSIONAL EXPERIENCE

Ericsson (China) Sep 2025 – Nov 2025
AI Engineering Intern

- Enhanced a Django Fusion-RAG platform with BM25, ChromaDB, and RRF; diagnosed ranking failures through retrieval evaluation.
- Built LLM-generated QA and LLM-as-a-judge evaluation with Hit Rate/MRR, raising MRR by 0.10.
- Decoupled retrieval and reranking inputs to reduce context usage; added HyDE query rewriting for forum search.
- Added DB refresh, upload isolation, and HITL review tools; briefed a senior manager in Ericsson Sweden on the full project.

Beijing Silu Innovation Technology Jun 2025 – Jul 2025
Back-End Development Intern

- Migrated enterprise web modules from Spring MVC to Spring Boot and Vue.
- Automated code-repository documentation through a reusable web scraper.

RESEARCH EXPERIENCE

Carnegie Mellon University Sep 2026 – Present
Research Assistant, Xu Lab | Multi-Agent Systems & VLA Safety for Autonomous Labs

University of Notre Dame Jul 2025 – Present
Research Intern, iSURE | Ongoing Remote Research Collaboration

SIGN: Shape-Aware Sperm Head Morphology Classification Aug 2025 – Present
Co-First Author | Ongoing Research

- Proposed INR shape priors for a frozen SAM3 classifier; reached 69.1% macro-F1 on SCIAN-PA and 98.7% on HuSHeM.
- Encoded masks into 256-D SIREN/DeepSDF latents and fused shape priors through FiLM and contrastive training.

D-GEM: Sparsely Supervised Surgical Video Segmentation Jul 2025 – Feb 2026
MICCAI 2026

- Co-designed DINOv3 Gated Transient and Evolving Anchor memories with reliability gating and cosine-similarity grouping.
- Method reached 74.2% mean mIoU on 3 benchmarks with 10 labeled frames/video, 10.3 points above re-prompted SAM3.

UKAST: KAN-Enhanced Swin Transformer for Medical Segmentation Aug 2025 – Oct 2025
IEEE ISBI 2026 | arXiv:2511.04084

- Refined GR-KAN replacement of Swin MLPs and implemented rational-function KAN components with coauthors.
- Ran most ablations in shared W&B experiments; gained 0.4 (2D) and 1.3 (3D) mean Dice points over SwinUNETR-RC.
- Led weekly KAN paper reviews and architecture discussions, turning findings into experiments and manuscript updates.

Chongqing University Sep 2025 – May 2026
Undergraduate Researcher

Dental Lesion Recognition
Undergraduate Thesis

- Fine-tuned YOLOv8-Seg on 3,616 tooth instances, attaining 0.782 mask mAP@0.5.
- Combined Panoramic Context, mask-guided FiLM, and Rank-MIL in Swin; reached 0.861 macro-F1 and 110 FPS.
- Built a web prototype with per-tooth scores and overlays; reported weekly experiments to my faculty advisor.

SKILLS

Languages: Python, C/C++, Java, SQL
ML / AI: PyTorch, JAX, scikit-learn, Hugging Face, Computer Vision, LLMs, RAG, Weights & Biases
Engineering: Git, Docker, Django, FastAPI, Spring Boot, Vue, ChromaDB, GCP, Spark, Kafka, Neo4j